Kishore J K

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.32
Thamarai International School, Thanjavur	Class XII (CBSE)	2023	90.20%
Chinmaya Vidhyalaya, Nagapattinam	Class X (CBSE)	2021	88.20%

PROJECTS

ECG Image to Time Series Converter

- Implemented an end-to-end AI pipeline for extracting 1D time-series data from realistic ECG images.
- Developed a PowerShell script using ECG generator kit to build training batches in realistic and B&W formats.
- Used Object Detection model (Detectron2 / Faster R-CNN) to detect and crops standard ECG leads.
- Used DL Segmentation model to isolate ECG signals from gridlines and background noise.
- Developed a PyTorch-based custom model using Cosine Annealing & Huber Loss to extract 1D time-series data.

Bio Inspired CNN for Digit classification

- Developed a biologically inspired CNN using handcrafted retinal and edge-detection filters.
- Trained the network for MNIST digit classification, achieving ~97% accuracy.
- Visualized intermediate activations across retinal, edge, and intersection-like processing stages.

Passive Shimming for MRI

- Reduced simulated MRI field inhomogeneity error by ~95% using optimization algorithms.
- Computed and optimized spherical harmonic shim coefficients up to 2nd order for a 1.5 T MRI system.
- Optimized passive shim-element placement within prescribed DSV to compensate spatial variations in the static field.

Infrared Thermography vs Oral Thermometry - Statistical Analysis

- Conducted a paired clinical-measurement study comparing FLIR and ICI infrared thermography with oral thermometry.
- Performed data preprocessing, normality testing, and paired t-test/Wilcoxon statistical analysis on Tmax measurements.
- Demonstrated significant temperature underestimation by infrared devices, indicating lack of clinical equivalence with oral thermometry.

Electric Grid Optimization

- Implemented Kruskal's algorithm with Union-Find to construct a minimum spanning tree for an electrical grid.
- Optimized node connections to minimize total transmission-line length while avoiding redundant connections.

RESEARCH INTERNSHIP

Summer Research Intern - PHCCO IISc Bangalore

Under Dr. Mohit Kumar Jolly, Department of Bioengineering, IISc.

- 40 days intensive hands on research program on Systems Biology and Mathematical modelling
- Modelled the spread of euchromatin accross different regions in genome of S. Pombe.
- Incorporated principles from bioinformatics, systems biology, and nonlinear dynamics.
- Did Group projects with people from different domains like Biology, Physics, Mathematics, Computation.

POSITIONS OF RESPONSIBILITY

Teaching Assistant - Linear Systems and Signals

- Assisted conducting Assignments, resolving student queries evaluating coursework.

EBSB - Cultural exchange and National integration club - Design Coordinator

- Coordinated Visual Design and branding of the club. Lead the team for event organisation

Sunshine - The councelling cell - Design and Multimedia Coordinator

- Designed annual newsletter and managed visual design and branding of the club.

Milan - Interhostel Sports and Cultural Meet - Design Coordinator

- Led the design team in developing creative and promotional materials for club events.

Alumni Club - Design and Multimedia Coordinator

- Developed creative and promotional material for club events

SKILLS

- Programming languages: C , C++ , Python
- Tools: Matlab, VS code
- DBMS: MySQL

RELEVANT COURSES

CSE: Algorithms and Data Structure Lab, Algorithms, Data Structure and Applications.

AI Related: Implementation of Natural Intelligence, Foundation of Natural Intelligence.

Maths Foundation: Probability and Random Variables, Elementary Linear Algebra, Statistical Inference in Bioengineering.

EXTRACURRICULAR

- Volunteer in NSS
- Organised unofficial cultural events in college level.
- Participated and won prizes in Art competitions.